Characterization of a Promising Mutant for Astaxanthin Bioprospecting in *Chromochloris zofingiensis*

Ethan Ashby[1,2], Tim Jeffers[2], Melissa Roth[2], Kris Niyogi[2], Clarissa Cheney[1]

[1]Department of Biology, Pomona College, Claremont, CA, 91711, [2]Howard Hughes Medical Institute, Department of Plant and Microbial Biology, University of California, Berkeley, CA 94720-3102

Abstract:

The ketocarotenoid astaxanthin is a valuable pharmaceutical compound and nutritional supplement due to its high antioxidant capacity. Microalgae are particularly promising and thoroughly-researched organisms for astaxanthin bioproduction. The microalgae *Chromochloris zofingiensis* accumulates ultrahigh cell densities and high astaxanthin content in glucose-supplemented media, making it a strong candidate for astaxanthin bioproduction. We identified that astaxanthin accumulation in wild type *C. zofingiensis* is mediated on the transcriptional level by reactive oxygen species (ROS) and nitrogen (nitrate) concentration. We characterized a novel *C. zofingiensis* mutant that displays better growth than wild type, elevated non-photochemical quenching (NPQ) capability, and a nitrogen-sensitive pigment profile redder than wild type. This mutant likely overaccumulates astaxanthin and may help elucidate N-mediated components of astaxanthin biosynthesis as well as targets to genetically engineer *C. zofinginesis* for astaxanthin bioprospecting.

Introduction:

- *C. zofingiensis* produces large quantities of astaxanthin when grown in glucose media
- High nitrogen rescues green pigmentation and attenuates astaxanthin production in WT cultures *(Fig 1)*
- Maximal astaxanthin production is achieved by high-light-induced ROS and limited nitrogen cultures

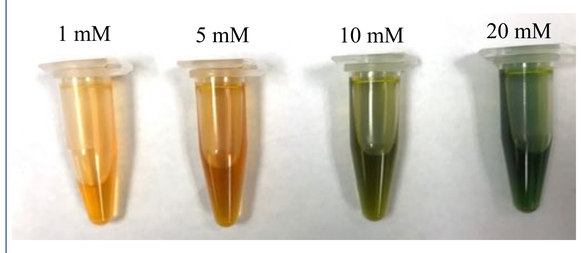

Figure 1: Acetone extracts of WT cultures grown in 10 mM glucose media with 1, 5, 10, 20 mM nitrate respectively. The reddish-orange pigmentation indicative of high astaxanthin accumulation in the lower nitrogen cultures.

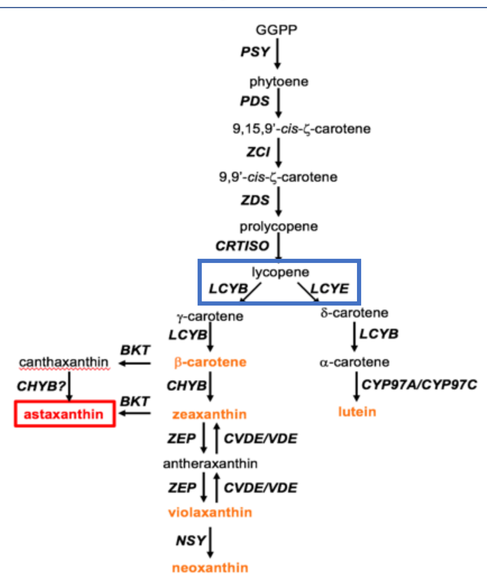

Figure 2: Carotenogenesis (carotenoid-producing) pathway with the LCYE/LCYB fork outlined in blue

The LCYE/LCYB fork: a control step?

- ROS upregulates *lcyb* and its downstream genes while downregulating *lcye* and its downstream genes
- *lcyb* and genes involved in the LCYB fork are generally transcriptionally upregulated in response to nitrogen stress
- *lcye* and genes involved in the LCYE fork are generally downregulated in response to nitrogen stress

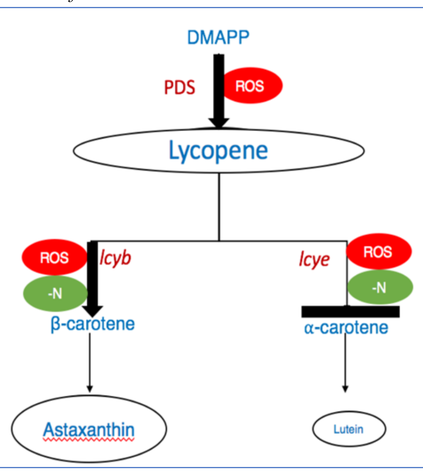

Figure 3: Schematic of proposed mechanism for astaxanthin biosynthesis

Proposed mechanism for astaxanthin biosynthesis

- Under ROS stress, genes involved early in the pathway are upregulated, increasing the lycopene pool size.
- ROS and low nitrogen downregulate *lcye*, generating **less** α-carotene and its downstream product, lutein.
- ROS and low nitrogen upregulate *lcyb*, generating **more** β-carotene and its downstream products, amongst them, astaxanthin.

Methods:

In order to better understand how carotenogenesis is regulated and leverage this knowledge for bioproduction, *C. zofingiensis* mutants that overaccumulate astaxanthin needed to be identified. To accomplish this, a forward genetic screen using a color-based colony screening approach was conducted *(Fig 4)*. Colonies that developed red pigmentation (indicative of high putative astaxanthin content) earlier than their neighboring colonies were selected. Cells were cultured in liquid media and phenotypically characterized to facilitate the construction of a library of mutants.

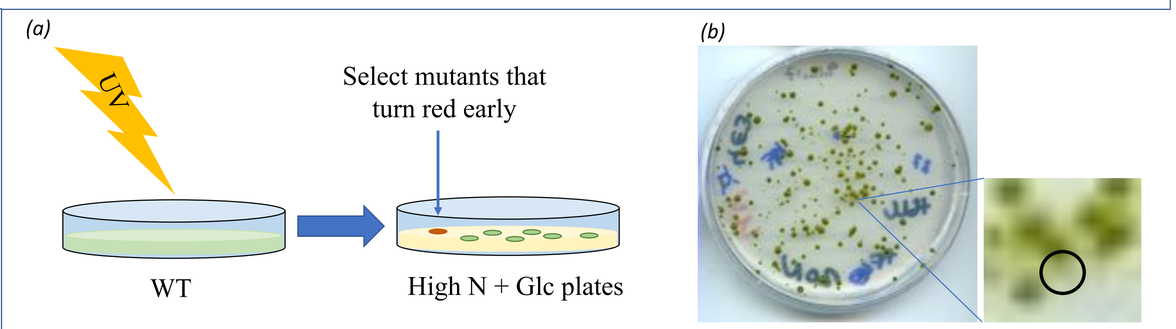

Figure 4: (a) UV mutagenesis protocol to identify astaxanthin overaccumulating mutants in C. zofingiensis (b) an example red colony that was selected for further characterization (outlined in black)

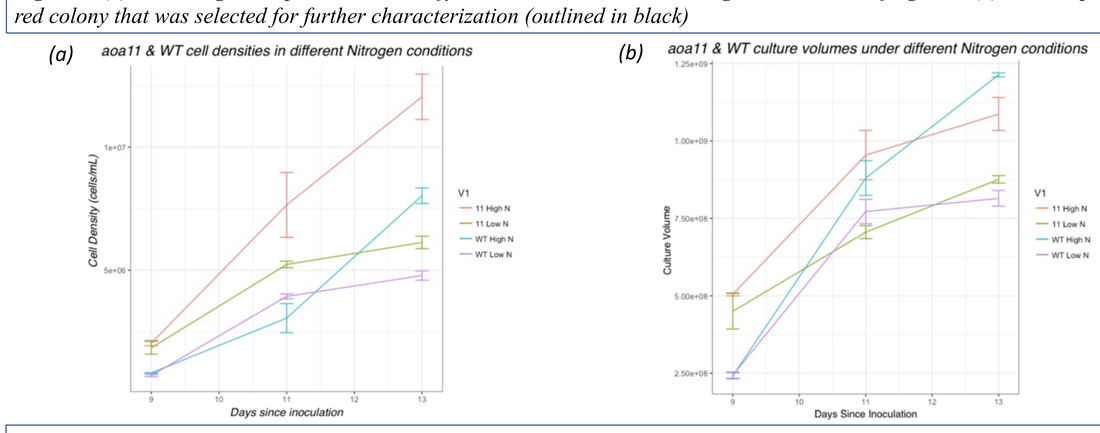

Figure 5: (a) aoa11 achieves significantly higher cell densities than WT under high (10 mM) & low (5 mM) N concentrations (b) aoa11 achieves comparable culture volumes to WT under high & low N conditions

Results:

- One particular mutant, *aoa11,* achieved higher cell densities and comparable culture volumes to the WT cultures in high and low nitrogen cultures supplemented with glucose *(Fig 5)*
- *aoa11* displayed elevated photoprotective (NPQ) capacity in comparison to the WT
- *aoa11* achieved an earlier and larger production of red pigment than the WT culture in high & low nitrogen cultures, indicative of astaxanthin overaccumulation *(Fig 6)*

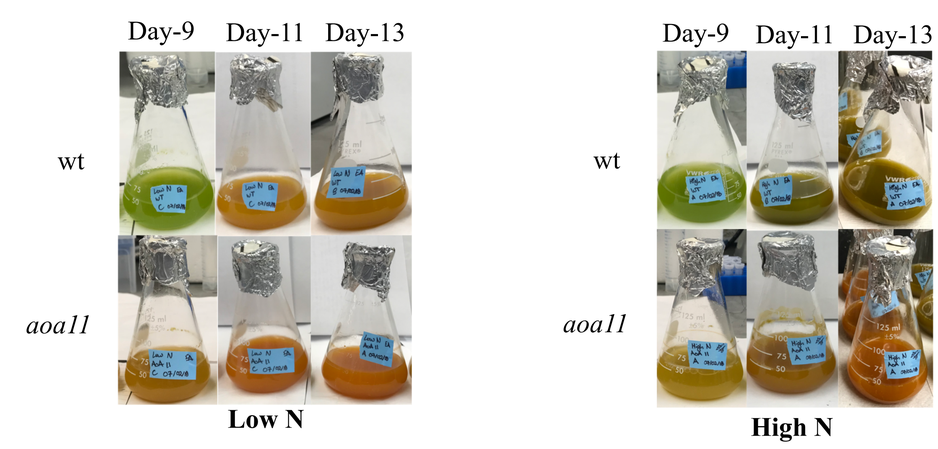

Figure 6: aoa11 achieves red pigmentation earlier and more intensely in both low and high nitrogen conditions

Conclusions and Next Steps:

- *aoa11* is a promising strain for astaxanthin bioproduction due to its high growth rates and putative astaxanthin overaccumulation in various culture conditions.
- aoa11's phenotypic characterization will facilitate the search for more mutants like *aoa11*; once a library of mutants is constructed, whole genome sequencing and SNP-mapping will ID the gene/genes responsible for this phenotype
- Hopefully, these findings will elucidate the roles N and ROS play in the transcription of genes involved in carotenogenesis and aid in bioengineering efforts in this organism moving forward.

Acknowledgments:

- A sincere thanks to the SURP donors who funded this project and made research aspirations a reality for me and others
- Thanks to Dean Tony Boston for organizing and chairing the SURP program
- A heartfelt thanks to Professor Cris Cheney for her guidance through all phases of the SURP process
- Thanks to the Niyogi Lab, particularly Tim Jeffers, Dr. Melissa Roth, and Professor Kris Niyogi for their thoughtful mentorship during my tenure at UC Berkeley